Strategic Production of Persistent Information: Finite-Population Incentives and Disappearing Common Jumps

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September 10, 2026
Independent working paper, Public v1
Developed independently with AI assistance; not peer reviewed.

Abstract

Persistent public information can benefit many users, yet its production depends on private incentives. We ask whether decentralized production can disappear as a population grows even when coordinated production remains useful. Under uniformly allocated, nontransferable opportunities and readiness costs incurred before assignment, an eligible player's effort raises discovery intensity by λ_M/N. In a six-state population with a continuous decaying information board, we construct a pure continuous Markov equilibrium on each finite horizon, unique within the bounded jointly continuous mild-HJB class. Without transfers, bounded private rewards imply that individual effort and aggregate artifact intensity are $O(N^{-1})$. Coupling to the system without artifacts then controls equilibrium paths and welfare. In a seeded, zero-board setting without routine deposits, an explicit productive region admits a fixed coordinated policy with a uniformly positive per-capita welfare gain. This policy retains common jumps and conditionally independent tagged paths whose task rewards have positive unconditional covariance. The comparison concerns a specified policy, without claiming convergence of optimal controls or values. A two-state benchmark isolates the role of access allocation, classifies productivity scalings, and admits a balanced creator prize implementing its planner. A two-agent counterexample shows why that implementation need not extend to heterogeneous private states. The results describe a conditional institutional comparison; fixed capacity alone is insufficient, and no empirical validity for multi-agent AI systems is established.

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1 Question, technology, and an exact benchmark

A reusable discovery can affect many people's later tasks. Its producer, however, pays a private cost and receives only its own share of the resulting benefit. This paper asks when that incentive wedge changes the large-population information environment itself: can consequential public artifacts disappear in equilibrium while a coordinated policy continues to produce them and raises average welfare?

The comparison holds the task dynamics and physical opportunity budget fixed. Opportunities are allocated uniformly among all players, cannot be reassigned, and require costly readiness before allocation. These assumptions give each eligible player's effort a marginal discovery rate λ_M/N. Bounded private rewards then limit willingness to prepare. Coordinated production internalizes the artifact's effect on average task rewards; in a specific productive region its benefit remains positive uniformly in population size. Assigning the same opportunity stream to a designated creator instead gives an N-independent private problem in the benchmark. Thus access allocation, as well as capacity, matters.

The main comparison connects four results: finite-population equilibrium existence, vanishing equilibrium artifact production, a strictly productive fixed policy, and different limiting dependence across users. The continuous board requires an explicit verification argument, retained in the appendix. The welfare comparison requires a positive adopter seed, a positive fraction of incomplete unexposed players, zero initial board, and no routine deposits. It establishes that artifacts can be useful in this region; it does not establish that randomness itself improves welfare.

An earlier nonstrategic note (Sneller, 2026) compares frequent small deposits with rare consequential artifacts. This companion endogenizes only the latter channel. All transitions, initialization assumptions and limit arguments needed for the present claims are stated here; the earlier note is separate background. Production has known productivity, observed states, and no Bayesian learning. Shared files or reusable procedures are hypothetical motivations rather than evidence for the opportunity institution.

The two-state benchmark below makes the economic distinction explicit. Section 2 constructs the finite game and equilibrium; Sections 3–4 establish the welfare and limiting comparison. Section 5 studies creator credit and its boundary, followed by related work, numerical evidence, and limitations.

1.1 Public information and admissible behavior

Let $Z \in \{0,1\}$ indicate useful-artifact availability. When $Z = 0$, player i chooses $u_i \in [0, \bar{u}]$ and the transition $0 \to 1$ has intensity $a_N \sum_i u_i$. When $Z = 1$, effort is required to be zero and $1 \to 0$ occurs at rate d. Initially $Z_0 = z$ is observed. Players observe the public state history, time, and their own controls. Current efforts are not an argument of rivals' strategies. A deviation preserves their prescribed rules. There is no unknown productivity or Bayesian posterior.

Throughout this section $\beta, \gamma > 0$, $d, v, \ell, a_N \geq 0$, and $\bar{u} < \infty$. The cost is $c(u) = \ell u + \gamma u^2/2$ and the payoff is

$$J_i(z) = \mathbb{E}_z \int_0^\infty e^{-\beta t}[vZ_t - c(u_i(t))]\,\mathrm{d}t.$$

Admissible deviations are predictable controls in the stated interval, using no future randomness. Intensities are bounded by $Na_N\bar{u} + d$ and discounted costs by $c(\bar{u})/\beta$, so all such controls are nonexplosive and integrable. The equilibrium concept is stationary Markov perfection; the verification below allows arbitrary admissible unilateral deviations. Uniqueness is claimed only within symmetric stationary Markov equilibria. A zero cap is the trivial game.

Write $k = \beta + d$ and $A_N = Na_N$. The normalized technology is $a_N = \lambda/N$; the fixed-productivity counterfactual is $a_N = \lambda$. These are different sequences of economic technologies, not two units for the same population experiment.

1.2 Solution with linear and quadratic costs and a cap

For a symmetric profile u, let $D = V_1 - V_0$. Its consistency equations are

$$\beta V_0 = -c(u) + A_N u D, \qquad \beta V_1 = v - dD, \qquad (k + A_N u)D = v + c(u). \tag{1}$$

A player's maximization treats the other $(N-1)$ efforts as fixed:

$$u = \operatorname{clip}_{[0,\bar{u}]}((a_N D - \ell)/\gamma). \tag{2}$$

This is a global maximizer because the Hamiltonian is strictly concave.

Proposition 1.1 (Complete symmetric solution)**.** *Assume $a_N > 0$ and $\bar{u} > 0$. If $a_N v \le k\ell$, the unique symmetric stationary equilibrium has $u_N = 0$ and $D_N = v/k$. Otherwise let $\widehat{u}_N > 0$ be the unique positive solution of*

$$\gamma(N - \tfrac{1}{2})u^2 + \left(\frac{k\gamma}{a_N} + (N-1)\ell\right) u = v - \frac{k\ell}{a_N}. \tag{3}$$

Then $u_N = \min(\bar{u}, \widehat{u}_N)$ and

$$D_N = \frac{v + c(u_N)}{k + A_N u_N}, \qquad V_{N,1} = \frac{v - dD_N}{\beta}, \qquad V_{N,0} = \frac{v - kD_N}{\beta}.$$

The all-zero profile is an equilibrium exactly when $a_N v \le k\ell$.

The average-welfare planner uses the same effort for all players in state zero. It is inactive exactly when $A_N v \le k\ell$. Otherwise its unconstrained effort is the positive root of

$$\frac{\gamma}{2}u^2 + \frac{k\gamma}{A_N}u = v - \frac{k\ell}{A_N}, \tag{4}$$

and $u_{\mathrm{SP}} = \min(\bar{u}, \widehat{u}_{\mathrm{SP}})$. Its values follow (1) with that effort.

Proof. An interior Nash choice gives $D = (\ell + \gamma u)/a_N$; substituting into (1) gives (3). Its left side is strictly increasing for $u \ge 0$. At the cap the Kuhn–Tucker condition is

$$a_N \frac{v + c(\bar{u})}{k + A_N \bar{u}} \ge \ell + \gamma \bar{u}.$$

After multiplication by the positive denominator this is equivalent to $\widehat{u}_N \ge \bar{u}$. At zero the reverse marginal inequality gives the stated threshold. This exhausts the three cases and proves symmetric uniqueness. For a fixed sum of efforts, Jensen's inequality makes equal efforts cost minimizing under average welfare; hence the planner maximizes $A_N u D - c(u)$. Replacing a_N by A_N in its marginal condition gives (4) and the cap rule.

To verify, for any unilateral predictable control x_t, apply the finite-state jump compensation formula to $e^{-\beta t}V_{Z_t}$ up to time H. The Bellman maximum implies

$$V_z \ge \mathbb{E}_z\left[\int_0^H e^{-\beta t}(vZ_t - c(x_t))\,\mathrm{d}t + e^{-\beta H}V_{Z_H}\right].$$

Equality holds at the candidate effort. Boundedness of V eliminates the last term as $H \to \infty$. The identical calculation, after Jensen's inequality, verifies the planner against all joint predictable controls. This proves more than a first-order condition. □

When $a_N = 0$ or $v = 0$, zero effort is optimal; when the cap is zero the inactivity condition is vacuous. The positive-root function used in the code is $2s/(b + \sqrt{b^2 + 4as})$ for $au^2 + bu = s > 0$, avoiding cancellation.

1.3 Normalized quadratic benchmark and asymptotic orders

Set $\ell = 0$ and $a_N = \lambda/N$, with $\lambda, v > 0$. In the interior,

$$kD_N + \frac{\lambda^2}{\gamma}\left(\frac{1}{N} - \frac{1}{2N^2}\right)D_N^2 = v, \qquad u_N = \frac{\lambda D_N}{\gamma N}, \tag{5}$$

$$kD_{\mathrm{SP}} + \frac{\lambda^2}{2\gamma}D_{\mathrm{SP}}^2 = v, \qquad u_{\mathrm{SP}} = \frac{\lambda D_{\mathrm{SP}}}{\gamma}. \tag{6}$$

Both are positive roots. The Nash effort is at most the planner effort: the Nash polynomial in u has larger positive coefficients than the planner polynomial and the same constant surplus. At $N = 1$ the solutions coincide. From the planner equation $\gamma u_{\mathrm{SP}}^2/2 < v$, so $\bar{u} \geq \sqrt{2v/\gamma}$ suffices for both solutions to be interior. For smaller caps Proposition 1.1 applies; every fixed positive cap eventually ceases to bind at Nash equilibrium.

Put $L = \lambda^2 v/(\gamma k)$. With fixed positive primitives and $d > 0$,

Quantity	Symmetric Nash	Coordinated optimum
Individual effort (absent)	$\lambda v/(\gamma k N) + O(N^{-2})$	positive constant
Total effort (absent)	$\lambda v/(\gamma k) + O(N^{-1})$	$N u_{\mathrm{SP}}$
Discovery hazard (absent)	$L/N + O(N^{-2})$	$\lambda u_{\mathrm{SP}} > 0$
Stationary event frequency	$L/N + O(N^{-2})$	$d\lambda u_{\mathrm{SP}}/(d + \lambda u_{\mathrm{SP}})$
Stationary occupancy	$L/(dN) + O(N^{-2})$	$\lambda u_{\mathrm{SP}}/(d + \lambda u_{\mathrm{SP}})$
Value from state 0	$\lambda^2 v^2/(\beta\gamma k^2 N) + O(N^{-2})$	$W_0 > 0$

Here the planner column allows a binding cap. For an interior planner $W_0 = \lambda^2 D_{\mathrm{SP}}^2/(2\gamma\beta)$. The stationary net flow reward at constant effort and hazard q is

$$\bar{r} = \frac{vq - dc(u)}{d + q}.$$

This is distinct from the discounted value at a specified initial state. Under normalized Nash it is $vL/(dN) + O(N^{-2})$; under an active planner it is positive and independent of N (the planner marginal equation gives $vq > dc(u)$, also in the binding-cap case).

1.4 Waiting times, occupancy, and welfare baselines

For any of the stationary policies, put $q = A_N u$. Starting absent, the first discovery time is exponential with mean $1/q$ and $\mathbb{P}_0(\tau \leq T) = 1 - e^{-qT}$, with mean infinity and probability zero if $q = 0$. Starting present, the time to a *new* discovery is the sum of independent exponentials of rates d and q. For $d, q > 0$ its mean is $1/d + 1/q$, and its survival function is

$$\mathbb{P}_1(\tau > T) = \begin{cases} (qe^{-dT} - de^{-qT})/(q - d), & q \neq d, \\ e^{-qT}(1 + qT), & q = d. \end{cases} \tag{7}$$

If either rate is zero there is no new discovery from state one. If an initial artifact counts as "first useful information," that waiting time is instead zero. The occupancy probability solves $p' = q - (q + d)p$; for $q + d > 0$, $p(t) = q/(q + d) + (z - q/(q + d))e^{-(q+d)t}$.

The planner dominates the Nash profile from either initial state. Under normalized quadratic effort the limiting additive gaps are

$$W_0 - V_{N,0} \longrightarrow \frac{v - kD_{\mathrm{SP}}}{\beta} = W_0 > 0, \qquad W_1 - V_{N,1} \longrightarrow \frac{d}{\beta}\left(\frac{v}{k} - D_{\mathrm{SP}}\right). \tag{8}$$

Adding a constant flow reward b shifts every value by b/β and leaves these differences unchanged, while altering any welfare ratio. The initial-state-one gap vanishes at $d = 0$: a permanent initial artifact already provides the full reward.

For each finite N with positive Nash effort, discovery from absence eventually occurs almost surely. At $d = 0$, any positive hazard gives long-run occupancy one, although fixed-horizon discovery still vanishes under normalized Nash. At $d = q = 0$ every mixture of the two absorbing states is stationary. None of these stationary claims permits interchanging $T \to \infty$ and $N \to \infty$. Infinite discounted payoffs require $\beta > 0$; the finite-horizon population results below also permit $\beta = 0$. Sending both β and d to zero with N is a different asymptotic problem. With fixed $\beta > 0$, letting $\lambda \downarrow 0$ or $v \downarrow 0$ continuously removes effort and the gap.

1.5 Positive opportunity cost and the counterfactual

For normalized discovery, the all-zero Nash condition is $\lambda v/(Nk) \le \ell$, while planner activity requires $\ell < \lambda v/k$. Thus for any fixed $\ell \in (0, \lambda v/k)$ the planner is active but equilibrium is exactly inactive for every integer $N \ge \lambda v/(k\ell)$. Equation (3) supplies the positive-effort cases including a binding cap. The exact threshold is not merely an asymptotic order estimate.

For fixed individual productivity $a_N = \lambda$, quadratic costs instead give

$$kD_N + \frac{\lambda^2}{\gamma}(N - \tfrac{1}{2})D_N^2 = v, \qquad u_N = \lambda D_N/\gamma \sim \sqrt{v/(\gamma N)}, \qquad q_N \sim \lambda\sqrt{vN/\gamma}.$$

Hence aggregate discovery grows, occupancy tends to one when $d > 0$, stationary event frequency tends to d, and both Nash and planner values from zero tend to v/β. The planner's individual effort tends to $\min(\bar{u}, \sqrt{2v/\gamma})$ and its hazard is order N. For fixed $\ell > 0$, Nash is inactive for all N if $\lambda v \le k\ell$; otherwise

$$u_N \sim \frac{v - k\ell/\lambda}{\ell N}, \qquad q_N \longrightarrow \lambda v/\ell - k > 0. \tag{9}$$

The planner becomes active once $N\lambda v > k\ell$ and its hazard again grows linearly for a positive cap. There is no aggregate collapse in either active fixed-productivity case. The linear-cost Nash limiting values are $V_0 = (v - k\ell/\lambda)/\beta$ and $V_1 = (v - d\ell/\lambda)/\beta$.

1.6 Fixed capacity and the allocation of access

The opportunity construction in Section 2 separates capacity from allocation. If each rate-$\lambda\bar{u}$ opportunity is independently assigned to i with exogenous probability π_i, and success probability is $u_i/\bar{u}$, the absent-state hazard is $\lambda \sum_i \pi_i u_i$. Uniform allocation gives $\pi_i = 1/N$ and the normalized technology. Keeping this same total opportunity stream but setting $\pi_1 = 1$ instead gives $q = \lambda u_1$; players without access choose zero costly readiness.

Proposition 1.2 (Designated access in the two-state benchmark)**.** *Fix $\lambda, v, \beta, \gamma, \bar{u} > 0$, $d, \ell \ge 0$. With all opportunities assigned to player 1, its unique stationary optimal effort is zero exactly when $\lambda v \le (\beta + d)\ell$. Otherwise it is*

$$u_D = \min(\bar{u}, \widehat{u}_D) > 0, \qquad \frac{\gamma}{2}\widehat{u}_D^2 + \frac{k\gamma}{\lambda}\widehat{u}_D = v - \frac{k\ell}{\lambda}, \qquad k = \beta + d.$$

Its private incentive and discovery rate $q_D = \lambda u_D > 0$ are independent of N.

Proof. The creator's Bellman equations give $(k+\lambda u_D)D = v + c(u_D)$ and $u_D = \mathrm{clip}_{[0,\bar{u}]}((\lambda D - \ell)/\gamma)$. These are exactly the one-player equations of Proposition 1.1, including its cap rule and verification. Other players have no effect on discovery and strictly prefer zero effort. □

Thus fixed capacity alone does not imply collapse. Concentrating access changes the institution and does not refute the uniform-allocation theorem. It neither derives endogenous hierarchy nor proves delegation efficient, stable, implementable, or first-best in the full model. In particular, the creator still values only its own reward; average welfare counts its cost as $c(u_D)/N$. Convex costs favor spreading a fixed total effort only when marginal productivities are equal. Under the original uniform allocation, restricting production to K_N agents without reallocating their access still gives $\Lambda_N \le \lambda_M \bar{u} K_N / N$.

1.7 A benchmark scaling classification

Proposition 1.3 (Conditional discovery hazard under productivity scaling)**.** *In the symmetric stationary two-state game let $a_N = \lambda N^{-s}$, $s \ge 0$, with fixed $\lambda, v, \beta, \gamma, \bar{u} > 0$, $d \ge 0$, and $\ell = 0$. Every fixed positive cap is eventually inactive. Let $D_c > 0$ solve $kD_c + \lambda^2 D_c^2/\gamma = v$. Then:*

Regime	*Individual effort u_N*	*Absent-state hazard $q_N = N a_N u_N$*
$0 \le s < 1/2$	$\sim \sqrt{v/\gamma}\, N^{-1/2}$	$\sim \lambda\sqrt{v/\gamma}\, N^{1/2-s} \to \infty$
$s = 1/2$	$\sim (\lambda D_c/\gamma) N^{-1/2}$	$\to \lambda^2 D_c/\gamma > 0$
$s > 1/2$	$\sim (\lambda v/(\gamma k)) N^{-s}$	$\sim (\lambda^2 v/(\gamma k)) N^{1-2s} \to 0$

For fixed $\ell > 0$, every $s > 0$ instead gives eventual exact inactivity. At $s = 0$ the active region is $\lambda v > k\ell$ and $q_N \to \lambda v/\ell - k > 0$, as in (9).

Proof. Without the cap the positive value-gap root satisfies

$$kD_N + \frac{\lambda^2}{\gamma}(N^{1-2s} - \tfrac{1}{2}N^{-2s})D_N^2 = v, \qquad u_N = \lambda N^{-s} D_N/\gamma.$$

For $s > 1/2$, $D_N \to v/k$; at $s = 1/2$, $D_N \to D_c$; for $s < 1/2$, $D_N \sim \sqrt{\gamma v}\, N^{s-1/2}/\lambda$ by dividing by the dominant quadratic coefficient. All three expressions give $u_N \to 0$, proving eventual cap inactivity and the table. With linear cost, the exact inactivity condition $\lambda N^{-s} v \le k\ell$ eventually holds for every $s > 0$. □

For $d > 0$, occupancy is $q_N/(d+q_N)$ and stationary event frequency is $dq_N/(d+q_N)$. A diverging absent-state hazard therefore gives occupancy tending to one and event frequency tending to d, not infinity. For $d = 0$, positive discovery leads to absorbing occupancy one and long-run event frequency zero. Zero productivity, benefit, or cap removes activity; discount is fixed strictly positive, so joint vanishing-discount limits are excluded. The chosen $s = 1$ lies within a broader incentive-collapse region. This is a benchmark classification, not a six-state theorem or a claim of a new phase transition. Large population size alone selects none of these regimes.

2 Controlled population and finite-N equilibrium

2.1 States, transitions, and controls

Let $Y_i = (X_i, C_i) \in S = \{0,1\} \times \{\mathsf{U}, \mathsf{E}, \mathsf{A}\}$, $\mu^N = N^{-1} \sum_i \delta_{Y_i}$, and $B^N \geq 0$. The uncontrolled transition rates are

$$\begin{array}{lll} (0,c) \to (1,c) : c(c), & (1,c) \to (0,c) : r, & c(\mathsf{U}) = c(\mathsf{E}) = c_0,\ c(\mathsf{A}) = c_1, \\ (x,\mathsf{U}) \to (x,\mathsf{E}) : \rho b, & (x,\mathsf{E}) \to (x,\mathsf{A}) : \alpha, & (x,\mathsf{E}) \to (x,\mathsf{U}) : \nu, \\ (x,\mathsf{A}) \to (x,\mathsf{U}) : \delta. & & \end{array}$$

All rates and board impacts are finite and nonnegative. The board decays at rate κ. If $w = 1$, routine posts occur at per-adopter rate λ_w with jump η/N; if $w = 0$ there are no routine posts. We distinguish λ_w from the toy model's λ and write λ_M for rare productivity. Only the rare channel is controlled:

$$\Lambda^N(t, \boldsymbol{y}, b) = \frac{\lambda_M}{N} \sum_{i=1}^{N} \mathbf{1}_{\{c_i = \mathsf{A}\}} u_i(t, \boldsymbol{y}, b), \qquad 0 \leq u_i \leq \bar{u}. \tag{10}$$

A rare event adds h to the board and changes no private state. Effort has no other dynamical or private benefit. Creator marks are irrelevant until transfers are introduced in Section 5.

One explicit institutional interpretation is a fixed stream of experimental opportunities at rate $\lambda_M \bar{u}$, each assigned uniformly among the N players. Effort represents costly readiness chosen before the assignment: an assigned eligible player succeeds with probability $u_i/\bar{u}$, and an ineligible assignment expires unused. Independent thinning gives exactly (10). Readiness costs accrue continuously, whether or not an opportunity arrives. The institution is a fixed opportunity budget with exogenously dispersed access, non-reassignable allocations, and readiness costs incurred before assignment. Fixed capacity bounds total proposals; uniform access makes an individual's marginal success rate λ_M/N; expiration prevents redirecting unused assignments to an eligible prepared player. Paying for effort only after assignment would change the cost calculation. The unit of readiness, its cap, and (ℓ, γ) stay fixed across N, as do per-person rewards; welfare is their average. This is not a change of units or a fixed total benefit divided among more people.

A facility issuing nontransferable lottery slots to users who must maintain readiness is an illustrative interpretation. Reassignable access, a designated operator, or preparation after allocation may make it a poor description. No evidence supplied here establishes these restrictions in the motivating AI-agent setting. Section 1.6 isolates allocation within the two-state benchmark. The zero-cap case has no opportunities and needs no division by $\bar{u}$.

For the full game all labels' states and the board are publicly observed. The state for arbitrary asymmetric feedbacks is $s = (\boldsymbol{y}, b) \in S^N \times \mathbb{R}_+$. The generator is

$$\begin{aligned} \mathcal{L}^u_N f(s) = & -\kappa b \partial_b f(s) + \sum_i \sum_{y'} q_{y_i y'}(b)[f(s^{i \to y'}) - f(s)] \\ & + w\lambda_w \sum_i \mathbf{1}_{\{c_i = \mathsf{A}\}}[f(\boldsymbol{y}, b + \eta/N) - f(s)] + \Lambda^N(s)[f(s^h) - f(s)]. \end{aligned} \tag{11}$$

For symmetric feedback $u_i = \theta(t, y_i, \mu, b)$, (μ, b) is Markov and its generator has individual part

$$-\kappa b \partial_b f + N \sum_{y,y'} \mu(y) q_{yy'}(b)[f(\mu + (\delta_{y'} - \delta_y)/N, b) - f(\mu, b)],$$

routine part $wN\lambda_w a(\mu)[f(\mu, b + \eta/N) - f(\mu, b)]$, and rare part

$$\lambda_M \sum_{y : C(y) = \mathsf{A}} \mu(y) \theta(t, y, \mu, b)[f(\mu, b + h) - f(\mu, b)].$$

Here $a(\mu) = \sum_{y:C(y)=\mathsf{A}} \mu(y)$. A tagged player's value needs (y_i, μ, b) even under symmetry. Its generator separates its own rate-q jump (changing both y_i and μ) from rival jumps with counts $N\mu(y) - \mathbf{1}_{\{y_i=y\}}$. An empirical law alone loses the tagged state; with asymmetric rules it can also lose the population's Markov property.

Fix $T < \infty$, $\beta \geq 0$, $\gamma > 0$, $\ell \geq 0$, $0 \leq \bar{u} < \infty$, and zero terminal values. For remaining-horizon discounting,

$$J_i(t, s; u) = \mathbb{E}_{t,s} \int_t^T e^{-\beta(r-t)} [g(Y_i(r)) - \mathbf{1}_{\{C_i(r)=\mathsf{A}\}} c(u_i(r))] \, \mathrm{d}r, \qquad 0 \leq g \leq R.$$

The substantive welfare result uses $g(x, c) = Rx$ and $c_1 > c_0$. Controls are predictable for the usual augmentation of the primitive-clock filtration, optionally enlarged by independent randomization. Future clock increments and proposal marks are unavailable when a control is chosen; ineligible effort is zero. For any such controls the board is dominated jointly by independent proposal counts:

$$\sup_{t \leq T} B_t^N \leq B_0^N + \frac{\eta}{N} P_w(N\lambda_w T) + h P_M(\lambda_M \bar{u} T). \tag{12}$$

The routine term is omitted at $w = 0$. After the finitely many board proposals, private total rate is bounded by N times a finite random constant. Interlacing constructs a nonexplosive adapted process, with unique law for a specified admissible control rule. Feedback gives the strong Markov property after time is included in the state. Costs are bounded, so integrability is automatic. The equilibria below are verified against predictable unilateral deviations. Rivals keep their state-feedback rules; their controls do not respond instantaneously to an observed deviating current effort.

2.2 Direct continuous-board existence and verification

The continuous board cannot be ignored in an equilibrium-existence argument. Here its unbounded exposure rate belongs entirely to an *uncontrolled* baseline process. The controlled perturbation has bounded rate and one board-jump operator. This special structure closes the finite-N problem.

Let P_r^0 denote the transition semigroup on $S^N \times \mathbb{R}_+$ with the rare channel disabled, retaining routine posts when $w = 1$. It is a positive contraction on bounded functions. It maps bounded continuous functions to bounded continuous functions and is jointly pointwise continuous in (r, s). To check this, restrict initial boards to a compact set and condition on routine proposal count. On each finite-count event the board stays bounded and individual rates are locally Lipschitz. A shared-clock coupling proves continuity; the Poisson tail permits dominated passage to unrestricted counts. This also gives stochastic continuity and compact containment locally uniformly in the initial board. On a bounded initial-board set and a bounded routine proposal count, the paths stay in a compact set. Continuous test functions are uniformly continuous there; the omitted Poisson tail costs at most twice their bound times its probability. This applies also to bounded jointly continuous tests $f(t, s)$ with varying time argument. Independent increments and the pathwise construction give the strong Markov restart property used in Appendix A.1. Global sup-norm strong continuity of P^0 is neither assumed nor needed.

For a vector of bounded continuous functions $V = (V_1, \ldots, V_N)$ define

$$\begin{aligned} z_i(s) &= V_i(s^h) - V_i(s), \qquad A_i[V](s) = \mathbf{1}_{\{c_i=\mathsf{A}\}} \operatorname{clip}_{[0,\bar{u}]} \left(\frac{(\lambda_M/N) z_i(s) - \ell}{\gamma} \right), \\ \Lambda[V](s) &= \frac{\lambda_M}{N} \sum_i A_i[V](s), \qquad L = \lambda_M \bar{u}, \\ G_i[V](s) &= g(y_i) - c(A_i[V](s)) + (L - \Lambda[V](s)) V_i(s) + \Lambda[V](s) V_i(s^h). \end{aligned} \tag{13}$$

Since $c(0) = 0$, this includes ineligible players. On $[a, b]$ with terminal vector ψ, the precise HJB notion used here is the killed-semigroup equation

$$V_i(t) = e^{-(\beta+L)(b-t)} P^0_{b-t} \psi_i + \int_t^b e^{-(\beta+L)(r-t)} P^0_{r-t} G_i[V(r)] \, \mathrm{d}r. \tag{14}$$

It requires no board derivatives. Equivalently, its un-killed version has discount β and driver $g_i - c(A_i[V]) + \Lambda[V]\Delta_h V_i$.

Theorem 2.1 (Finite-N continuous Markov equilibrium)**.** *For every finite N, T and the above primitives, (14) with terminal zero has a unique bounded jointly continuous solution on $[0, T] \times S^N \times \mathbb{R}_+$. Its feedback $A[V]$ is a pure continuous Markov-perfect equilibrium, verified against bounded predictable unilateral deviations. Values are unique within this bounded continuous mild-HJB class; controls are its unique Hamiltonian maximizers. Permutation invariance makes the constructed equilibrium symmetric. No compact-board truncation, classical derivatives, relaxed controls, or small-total-horizon restriction is imposed. Uniqueness outside this verification class or among history-based equilibria is not asserted.*

Proof. Use the maximum over indices and states as the vector norm. On $\|V\|, \|W\| \le M$, with $d_* = \|V - W\|$,

$$\|A_i[V] - A_i[W]\| \le \frac{2\lambda_M}{\gamma N} d_*, \qquad \|\Lambda[V] - \Lambda[W]\| \le \frac{2\lambda_M^2}{\gamma N} d_*.$$

Cost is $(\ell + \gamma\bar{u})$-Lipschitz on the effort interval. The weights $L - \Lambda$ and Λ are nonnegative and sum to L. Thus a Lipschitz constant for G on this ball is

$$C_M = L + \frac{4M\lambda_M^2 + 2\lambda_M(\ell + \gamma\bar{u})}{\gamma N}, \tag{15}$$

and $\|G[V]\| \le R + c(\bar{u}) + LM$. Both can be bounded independently of N. Let $K_t = R\int_0^t e^{-\beta r} \, \mathrm{d}r$, set $M = K_T + 1$, and take continuous terminal values in $[0, K_T]$. Work in the closed radius-M ball of $C_b([a, b] \times S^N \times \mathbb{R}_+)^N$ with terminal slice ψ, using the supremum norm; this is complete. Put $B_M = R + c(\bar{u}) + LM$ and let $\widehat{C}_M$ be (15) with $1/N$ replaced by 1. A slab length

$$b - a \le \min\{[2(B_M + 1)]^{-1}, [2(\widehat{C}_M + 1)]^{-1}\}$$

makes the operator norm at most $K_T + 1/2 < M$ and its contraction factor less than $1/2$. It preserves the terminal slice. Compact containment and bounded dominated convergence give joint continuity, including at the endpoints. Banach iteration gives a local solution with a slab length independent of N; no bound uniform over unbounded T is asserted.

Verification supplies the bound needed to continue. For a local solution use the un-killed driver $F_i = g_i - c(A_i) + \Lambda[A]\Delta_h V_i$. The mild identity and the Markov property give the baseline martingale

$$\begin{aligned} &e^{-\beta(r-t)} V_i(r, S^0_r) + \int_t^r e^{-\beta(s-t)} F_i(s, S^0_s) \, \mathrm{d}s \\ &\quad = \mathbb{E}\left[e^{-\beta(b-t)} \psi_i(S^0_b) + \int_t^b e^{-\beta(s-t)} F_i(s, S^0_s) \, \mathrm{d}s \mid \mathcal{F}_r \right]. \end{aligned}$$

Appendix A.1 proves killed/un-killed equivalence and the interlacing identity at random proposal times with predictable acceptance. In particular, under an actual unilateral deviation,

$$e^{-\beta(r-t)} V_i(r, S_r) + \int_t^r e^{-\beta(s-t)} [F_i(s, S_{s-}) - \Lambda^{\text{actual}}_s \Delta_h V_i(s, S_{s-})] \, \mathrm{d}s$$

is a true martingale. This uses baseline restarts and compensation, without assuming differentiability of V_i.

Relative to the candidate, a deviator's instantaneous residual is

$$\frac{\lambda_M}{N}(u_i - A_i)\Delta_h V_i - c(u_i) + c(A_i) \le 0,$$

since A_i is the global maximizer of the strictly concave Hamiltonian. Consequently V_i dominates every deviating payoff with terminal ψ_i, with equality at the candidate. Starting with zero terminal payoff and pasting backward verifies every constructed interval. At the next slab's terminal time b, the already verified continuation obeys $0 \le \psi_i \le K_{T-b}$. Zero effort guarantees nonnegative reward and continuation under fixed rival feedback. Nonnegative costs give the upper bound $K_{b-t} + e^{-\beta(b-t)}K_{T-b} = K_{T-t}$:

$$0 \le V_i(t,s) \le K_{T-t} \le K_T. \tag{16}$$

Hence the same ball and interval length can be used repeatedly. Finitely many steps cover $[0,T]$. The pasted solution satisfies the global mild identity by the semigroup property. Two bounded solutions lie in some common finite ball; contraction uniqueness on successive short slabs makes them identical. Finally, label permutations preserve all equations, proving symmetry. □

This is a class-specific existence result. It works because the unique local maximizer depends only on that player's jump-value difference: no unresolved simultaneous action selection is hidden in the coupled driver.

2.3 Uniform incentives, including asymmetric verified equilibria

Corollary 2.2 (Small private incentives)**.** *Every Markov equilibrium with bounded continuation values satisfying the local verification identity obeys*

$$u_i^{\mathrm{NE}} = \mathrm{clip}_{[0,\bar{u}]}\left(\frac{(\lambda_M/N)\Delta_h V_i - \ell}{\gamma}\right) \le \min\left(\bar{u}, \frac{(\lambda_M K_T/N - \ell)_+}{\gamma}\right) \tag{17}$$

at eligible states. In particular, for $\ell = 0$,

$$u_i^{\mathrm{NE}} \le \frac{\lambda_M K_T}{\gamma N}, \qquad \Lambda_N^{\mathrm{NE}} \le \frac{\lambda_M^2 K_T}{\gamma N} =: \frac{C_*}{N}.$$

For $\ell > 0$, all equilibrium exploration is zero whenever $\lambda_M K_T/N \le \ell$. This is sufficient, not a full characterization of activity below that threshold. Symmetry is unnecessary for the estimate.

Proof. The zero-control deviation gives the lower bound in (16); nonnegative costs give the upper bound. Thus $\Delta_h V_i \le K_T$. Insert this in the Hamiltonian maximizer and sum over at most N eligible players. The tighter bound K_{T-t} is available at time t. □

An artifact may change many future transitions and rivals' future feedback responses. All those effects are inside $V_i(s^h) - V_i(s)$, whose positive part cannot exceed the player's bounded total reward. No uniform board derivative or weak amplification hypothesis is required. Unbounded transfers and direct private effects of effort invalidate the premises.

2.4 The seed and the finite planner

If $B_0 = 0$ and all agents are unexposed, with no exposed or adopter seed, the communication system remains unexposed under every policy, also in hybrid mode. There is nobody eligible to produce the board needed for exposure. Task completion at c_0 can continue; it is the communication configuration that is absorbing.

For reference, the finite planner's formal empirical-state equation is

$$0 = \partial_t W_N + \mathcal{L}_N^0 W_N - \beta W_N + \langle \mu, g \rangle + a(\mu) \max_{0 \le u \le \bar{u}} \{\lambda_M u \Delta_h W_N - c(u)\}. \tag{18}$$

For a fixed total eligible effort, equal eligible effort minimizes cost and leaves the state law unchanged. Thus this scalar maximization is valid despite different private reward states. Its maximizer is $\mathrm{clip}_{[0,\bar{u}]}((\lambda_M \Delta_h W_N - \ell)/\gamma)$. The same bounded-perturbation argument gives a unique continuous bounded mild solution and a verified finite-N optimal feedback: the scalar maximized Hamiltonian is Lipschitz in its jump difference with constant $\lambda_M \bar{u}$; verification gives $0 \le W_N \le K_T$ and continuation. This establishes finite-horizon existence, not an explicit optimal solution or convergence of optimal policies. The following welfare proof needs only the planner supremum.

3 A uniform, strictly positive welfare improvement

3.1 Performance difference with an explicit remainder

Let $W_N^0(t, \mu, b)$ be average reward under zero exploration, with the same routine channel as the comparison system. It is a bounded baseline potential. Let eligible agents exert $\varepsilon\chi(t, \mu, b)$, where $0 \le \chi \le 1$ is Borel and predictable when evaluated at left limits, and $0 \le \varepsilon \le \bar{u}$. Let W_N^ε be the resulting welfare from the given initialization, including exploration costs. Write

$$A_N = \mathbb{E}^0 \int_0^T e^{-\beta t} a(\mu_t^N) \chi(t, \mu_t^N, B_t^N) [\lambda_M \Delta_h W_N^0(t, \mu_t^N, B_t^N) - \ell] \, \mathrm{d}t. \tag{19}$$

Theorem 3.1 (Uniform policy variation)**.** *For every N and any such gate,*

$$W_N^\varepsilon - W_N^0 = \varepsilon A_N + \mathcal{R}_N(\varepsilon), \qquad |\mathcal{R}_N(\varepsilon)| \le \left[2\lambda_M(\lambda_M K_T + \ell) T^2 + \frac{\gamma T}{2} \right] \varepsilon^2. \tag{20}$$

The same bound applies in rare-only and hybrid mode. No monotonicity or regularity of the value as a function of the board is needed for this identity.

Proof. The baseline potential identity, interlaced at bounded-rate artifact proposals as in Theorem 2.1, gives the exact performance difference

$$W_N^\varepsilon - W_N^0 = \mathbb{E}^\varepsilon \int_0^T e^{-\beta t} \left\{ \varepsilon a \chi (\lambda_M \Delta_h W_N^0 - \ell) - \frac{\gamma \varepsilon^2}{2} a \chi^2 \right\} \mathrm{d}t. \tag{21}$$

Terminal values are zero. This accounts for the current eligible fraction, the λ_M discovery productivity, subsequent reward gain, discounting, and both cost components.

Couple the controlled and baseline systems with identical initial data and private/routine clocks. Add an independent artifact proposal process of rate $\lambda_M \varepsilon$, accepting with probability $a\chi$. Before the first accepted artifact all states coincide exactly. The probability of any accepted artifact is at

most $\lambda_M \varepsilon T$. The integrand $Q = a\chi(\lambda_M \Delta_h W_N^0 - \ell)$ is bounded in absolute value by $M = \lambda_M K_T + \ell$. Therefore

$$\left| \mathbb{E}^{\varepsilon} \int_0^T e^{-\beta t} Q_t \, \mathrm{d}t - \mathbb{E}^0 \int_0^T e^{-\beta t} Q_t \, \mathrm{d}t \right| \leq 2MT \, \lambda_M \varepsilon T.$$

The quadratic cost term is bounded by $\gamma \varepsilon^2 T/2$. Substitution proves (20). The number of accepted events is dominated by $P(\lambda_M \varepsilon T)$, and $\mathbb{P}(J_T \geq 2) \leq (\lambda_M \varepsilon T)^2/2$ by its second factorial moment. Thus multiple artifacts are accounted for rather than suppressed in the first-order formula. □

3.2 A productive region with an explicit witness

Take rare-only mode, $B_0 = 0$, $\bar{u}, R, \rho, \alpha, h, \lambda_M > 0$, $c_1 > c_0$, and fixed finite nonnegative remaining rates. Initial deterministic counts satisfy

$$a(\mu_0^N) \geq a_* > 0, \qquad \mu_0^N(0, \mathsf{U}) \geq m_* > 0. \tag{22}$$

Counts may be randomly permuted independently of future clocks. Choose $0 < t_0 < T$ and $\chi = \mathbf{1}_{[0,t_0)}$. Endpoint choice has no effect on payoffs.

Lemma 3.2 (Rare-only zero-board monotonicity)**.** *With all future artifact production disabled, a forced board jump from zero at time t weakly increases expected task reward from every private initial state. For a player in $(0, \mathsf{U})$ at that time and $t \leq t_0$, its remaining value increase is at least $L_* > 0$, uniformly in N and in the other players' states.*

Proof. Without a board, U is absorbing for the communication coordinate. Couple baseline and forced-artifact communication chains identically until their first entry into U. This entry is simultaneous because rates out of E, A do not depend on the board. Thereafter the baseline stays U. Before that entry the two task completion intensities agree; afterwards the forced-artifact chain's completion intensity is always either c_0 or c_1, while the baseline's is c_0.

Use a common rate-c_0 clock that sets incomplete tasks to complete, a common rate-r clock that resets completed tasks, and an independent rate-$(c_1 - c_0)$ completion clock accepted when the corresponding communication state is A. The task coordinate in the forced-artifact chain remains at least the baseline coordinate. Communication states themselves need not be globally ordered. This proves nonnegative reward differences for every initial state.

For strictness put $s_* = (T - t_0)/4$, $b_* = he^{-\kappa T}$, and

$$\begin{aligned} q_* &= e^{-(c_0+r+\nu+\delta)T}(1 - e^{-\rho b_* s_*})(1 - e^{-\alpha s_*})(1 - e^{-(c_1-c_0)s_*}), \\ L_* &= Rs_* e^{-\beta T} q_* > 0. \end{aligned} \tag{23}$$

Start both chains at $(0, \mathsf{U})$ immediately after the forced artifact time. Forbid all potential c_0 completion, reset, forgetting, and abandonment clocks on the following four slots of length s_*. This has probability at least $e^{-(c_0+r+\nu+\delta)T}$; these are independent primitive Poisson clocks, including their rings when their transition is disabled. Require at least one rate-ρb_* exposure proposal in slot one, one adoption-clock ring in slot two, and one extra-completion ring in slot three. Exposure is feasible because the decaying forced board remains at least b_* throughout these slots. If a required favorable transition has already occurred, the forbidden clocks ensure it persists, so additional favorable rings do not invalidate the witness. The three independent positive probabilities multiply as in (23). The forced task is complete and the baseline task incomplete throughout slot four, yielding discounted reward at least $Rs_* e^{-\beta T}$. All other reward differences are nonnegative by the coupling. □

The witness uses independent potential clocks, including disabled rings, and the absorbing zero-board baseline. It supplies no global order of communication states at arbitrary positive boards.

Theorem 3.3 (Uniform coordination gain)**.** *Under the preceding productive-region assumptions, including* $\bar{u} > 0$*, (22), rare-only mode and* $\ell = 0$*,*

$$A_N \geq A_* := \lambda_M L_* a_* m_* \int_0^{t_0} e^{-(\beta+\delta+c_0)t}\,\mathrm{d}t > 0 \tag{24}$$

for every admissible population size. Write $C = 2\lambda_M^2 K_T T^2 + \gamma T/2$ *and choose* $0 < \varepsilon \leq \min(\bar{u}, A_*/(2C))$*. Then*

$$W_N^{\mathrm{SP}} \geq W_N^{\varepsilon} \geq W_N^0 + \delta_*, \qquad \delta_* = \varepsilon A_*/2 > 0.$$

For the equilibrium from Theorem 2.1, and for any other verified Markov equilibrium satisfying Corollary 2.2,

$$\liminf_{N\to\infty}(W_N^{\mathrm{SP}} - W_N^{\mathrm{NE}}) \geq \delta_*.$$

Proof. Along the zero-exploration law the board is zero and agents are independent given their deterministic initial labels. By Lemma 3.2, $\Delta_h W_N^0(t,\mu,0) \geq L_*\mu(0,\mathsf{U})$ for $t \leq t_0$. Let I be the set of initial adopters and J the disjoint set of initial $(0,\mathsf{U})$ players. An $i \in I$ that has no abandonment ring before t remains an adopter, with probability $e^{-\delta t}$. A $j \in J$ with no baseline completion ring remains $(0,\mathsf{U})$, with probability $e^{-c_0 t}$. Their clocks are independent. Consequently

$$\mathbb{E}^0[a(\mu_t^N)\mu_t^N(0,\mathsf{U})] \geq \frac{|I||J|}{N^2} e^{-(\delta+c_0)t} \geq a_* m_* e^{-(\delta+c_0)t}.$$

Insert this in (19); random permutation of the initial groups does not change the argument. The variation bound gives $W_N^{\varepsilon} - W_N^0 \geq \varepsilon A_* - C\varepsilon^2 \geq \delta_*$. The final inequality uses the equilibrium baseline convergence proved next. The planner supremum exceeds the feasible policy whether or not one uses its finite-N existence result. □

With linear costs, the same region gives $A_N^{(\ell)} \geq A_* - \ell t_0$, so a sufficiently small positive $\ell < A_*/t_0$ retains a strictly positive coordinated gain. The constant in (20) then includes ℓ. This is conservative but concrete, and large-N equilibrium is exactly inactive.

The proof does not assert positive planner effort at every state or close to the terminal time. It fails without a seed, without task benefit, or if the linear marginal cost is too large. For positive initial boards and for hybrid posting, (20) remains valid; the positivity theorem does not automatically carry over. A fresh positivity argument or the explicit condition $\liminf A_N > 0$ is needed there.

4 Disappearance versus a common-jump limit

4.1 Equilibrium: probability, strong coupling, and welfare

Set $\ell = 0$ initially and let $C_* = \lambda_M^2 K_T/\gamma$. The artifact count at any verified equilibrium satisfies

$$\mathbb{P}(J_T^{N,\mathrm{NE}} \geq 1) \leq \mathbb{E}J_T^{N,\mathrm{NE}} \leq C_* T/N. \tag{25}$$

Couple the equilibrium process to its rare-disabled version with identical initial state and identical individual and routine driving clocks. Effort changes no other transition, so all states agree exactly until the first rare artifact, even if rivals' future feedback reactions are highly state dependent. If Φ

is any bounded functional of the whole population path (including any fixed collection of tagged paths), then

$$|\mathbb{E}\Phi(S^{N,\mathrm{NE}}) - \mathbb{E}\Phi(S^{N,0})| \le 2\|\Phi\|_\infty C_* T/N. \tag{26}$$

This is a bounded path-test estimate; it is not by itself a bound for the unbounded board norm.

Proposition 4.1 (Strong extra error and welfare convergence)**.** *Suppose $B_0^N \le K$ almost surely. There is a coupling such that, in both rare-only and hybrid modes,*

$$\mathbb{E}\sup_{t\le T}(\|\mu_t^{N,\mathrm{NE}} - \mu_t^{N,0}\|_1 + |B_t^{N,\mathrm{NE}} - B_t^{N,0}| + J_t^{N,\mathrm{NE}}) \le C_T/N.$$

Writing average equilibrium and no-exploration welfare as W_N^{NE}, W_N^0,

$$|W_N^{\mathrm{NE}} - W_N^0| \le K_T C_* T/N + \frac{\lambda_M^2 K_T^2 T}{2\gamma N^2}. \tag{27}$$

For $\ell > 0$, all these differences are exactly zero for sufficiently large N.

Proof. Construct a common routine proposal process of rate $N\lambda_w$ with uniform label marks and an independent rare proposal process P_N of rate C_*/N. The first is valid for both populations; accept routine proposals when the marked agent is eligible. The second dominates the equilibrium rare intensity. Let $E_N = \{P_N(T) \ge 1\}$. Off E_N, the coupled paths coincide. On E_N, empirical error is at most 2, and joint board domination gives

$$\sup|B^{N,\mathrm{NE}} - B^{N,0}| \le 2K + \frac{2\eta}{N}P_w(N\lambda_w T) + hP_N(T).$$

The routine proposal count is independent of E_N. Therefore multiplication by $\mathbf{1}_{E_N}$ and expectation bounds the right side by $[2K + 2\eta\lambda_w T]\mathbb{P}(E_N) + hC_*T/N$. This proves the board estimate without a Hölder loss; the empirical and count estimates are immediate. The argument works with initial random boards bounded by K, without requiring independence of the initial populations from each other in this comparison. All future proposal clocks are independent of initial data.

Each player's total discounted task reward lies in $[0, K_T]$; on the coupling it can differ only on E_N. Average quadratic effort cost is at most $\gamma T(\lambda_M K_T/(\gamma N))^2/2$. Combining the two yields (27). If $\ell > 0$, Corollary 2.2 disables the rare channel exactly for large N, and effort cost is also zero. □

For a limiting statement impose the following initialization conditions: for some $p > 2$, $0 \le B_0^N \le K$ almost surely and

$$\varepsilon_{N,p} = \left\|\|\mu_0^N - \mu_0\|_1 + |B_0^N - b_0|\right\|_{L^p} \longrightarrow 0,$$

with deterministic (μ_0, b_0) and initial data independent of future clocks. The rare-disabled limit is

$$\dot\mu_t = F(\mu_t, b_t), \qquad \dot b_t = -\kappa b_t + w\eta\lambda_w a(\mu_t), \qquad F(\mu, b) = \sum_{y,y'}(\delta_{y'} - \delta_y)\mu(y)q_{yy'}(b). \tag{28}$$

The baseline argument in Appendix A.4, with $w = 0$ or 1, and Proposition 4.1 give expected uniform error $C_T(\varepsilon_{N,p} + N^{-1/2})$ to this deterministic system. For completeness, the baseline argument writes empirical and routine-board martingales of L^p supremum $O(N^{-1/2})$ and uses $|b^N\mu^N(y) - b\mu(y)| \le |b^N - b| + b^N\|\mu^N - \mu\|_1$. Pathwise Gronwall followed by Hölder is legitimate because

$$\mathbb{E}e^{q\eta P_w(N\lambda_w T)/N} = \exp\{N\lambda_w T(e^{q\eta/N} - 1)\}$$

is uniformly bounded for every fixed q.

Under exchangeable chaotic initialization, fixed tagged families converge in $D([0,T],S)^k$ to independent chains with rates $q_{yy'}(b_t)$. Before a tag's first discrepancy, the width of its exposure-clock mismatch is at most $\rho|B^N - b|$; the other rates agree when its own state agrees. Integration, the board estimate, and a union bound prove path coupling. Hybrid equilibrium therefore removes *rare* common jumps and leaves the deterministic weak background. Common random initialization or unrelated imposed common noise would remain; in that setting product laws must be conditioned on it.

4.2 Coordination under a fixed feedback

Fix, independently of N, eligible feedback $\theta(t,y,\mu,b) \in [0,\bar{u}]$, zero when $C(y) \neq \mathsf{A}$. Assume it is Borel in time and uniformly Lipschitz in (μ,b), with the same Lipschitz constant for all t,y. Finite deterministic time discontinuities, including the early gate, are allowed. Then

$$\mathcal{A}(t,\mu,b) = \lambda_M \sum_{y:C(y)=\mathsf{A}} \mu(y)\theta(t,y,\mu,b)$$

is bounded by $L = \lambda_M \bar{u}$ and globally Lipschitz in (μ,b). On a shared PRM $\Pi^0(\,\mathrm{d}t,\,\mathrm{d}z)$ of intensity $\mathrm{d}t\,\mathrm{d}z$, the candidate limit is

$$\begin{aligned} \dot{\mu}_t &= F(\mu_t, B_t), \\ \mathrm{d}B_t &= \{-\kappa B_t + w\eta\lambda_w a(\mu_t)\}\,\mathrm{d}t + h\,\mathrm{d}J_t, \\ J_t &= \int_{(0,t]\times[0,L]} \mathbf{1}_{\{z\le \mathcal{A}(s,\mu_{s-},B_{s-})\}} \Pi^0(\,\mathrm{d}s,\,\mathrm{d}z). \end{aligned} \tag{29}$$

It has a unique pathwise construction: solve the locally Lipschitz deterministic flow between finitely many proposals and decide acceptance from the left limit. The empirical law is continuous even though the common board jumps.

Theorem 4.2 (Fixed-policy convergence)**.** *Under the preceding feedback and initialization assumptions, in rare-only mode the finite system using the same feedback and common PRM satisfies*

$$\mathbb{E}\sup_{t\le T}(\|\mu_t^N - \mu_t\|_1 + |B_t^N - B_t| + |J_t^N - J_t|) \le C_T(\varepsilon_{N,p} + N^{-1/2}).$$

In hybrid mode the same coupled triple converges in expected uniform distance under these assumptions. The localization proof given here does not assert a quantitative hybrid rate for the general controlled feedback. With exchangeable chaotic initialization, for bounded Lipschitz path functionals $\Phi_1,\dots,\Phi_k$,

$$\mathbb{E}\left|\mathbb{E}\left[\prod_{j=1}^k \Phi_j(Y_j^N) \mid \mathcal{F}_T^0\right] - \prod_{j=1}^k \int \Phi_j(y)\mathcal{P}^0(\,\mathrm{d}y)\right| \longrightarrow 0, \tag{30}$$

where $\mathcal{F}_T^0 = \sigma(\Pi^0|_{[0,T]\times[0,L]})$ and $\mathcal{P}^0$ is the conditional law of a chain driven by independent private clocks in B.

Proof. For rare-only mode, condition on the number m and times of common proposals, but not on their independent uniform height marks. Both boards are bounded by $K + hm$. The private martingale has conditional L^p supremum bounded by $C(1+m)^q/\sqrt{N}$, using its bracket bound $C(1+K+hm)T/N$ and its $O(1/N)$ jump sizes. Conditioning on the proposal schedule leaves the initial error distribution unchanged. Private PRMs retain their intensities in the filtration enlarged

by that schedule, and height marks are revealed only at their proposals. Stop just before the first acceptance mismatch. Up to this time, common jumps agree and the drift comparison yields a Gronwall factor at most $e^{C(1+m)T}$. At any proposal the conditional mismatch probability is exactly $|\mathcal{A}^N - \mathcal{A}|/L$, bounded by a fixed constant times the pre-proposal state error. A union bound over the m marks thus gives conditional first-mismatch probability at most $C(1+m)^q e^{cm}(\varepsilon_{N,p} + N^{-1/2})$. After a mismatch the population error is at most 2, and the board/count error at most a constant times $1+m$. Poisson exponential moments sum this bound. The case $L = 0$ is the baseline LLN.

For hybrid convergence, stop both systems when either board first exceeds H. On the stopped domain all drift coefficients have a deterministic Lipschitz constant C_H and the compensated mismatch identity is

$$\mathbb{E}K^N_{t\wedge\tau_H} = \mathbb{E}\int_0^{t\wedge\tau_H} |\mathcal{A}(s,\mu^N_s,B^N_s) - \mathcal{A}(s,\mu_s,B_s)|\,\mathrm{d}s.$$

Private and routine martingale suprema tend to zero in L^p. The stopped drift, board, and count differences close under ordinary expectation and Gronwall. Hence the stopped error tends to zero for fixed H. Domination (12) gives uniform exponential moments, so the exit probability tends to zero uniformly as $H \to \infty$ and the full errors are uniformly integrable. Taking $N \to \infty$ first, then $H \to \infty$, proves expected uniform convergence. It gives no declared root-N constant after localization.

For tags, couple independent limiting chains to the finite tags with shared private PRMs and coupled initial states. Before a mismatch, only the exposure rate differs, by $\rho|B^N_s - B_s|$. Expected integrated board error and initial tag mismatch tend to zero. Given $\mathcal{F}^0_T$, the limiting initial samples and private PRMs are independent, so their product law is $(\mathcal{P}^0)^{\otimes k}$. Conditional Jensen and the telescoping product bound give (30). □

4.3 The retained noise has a productive tagged effect

Use the early policy $\theta = \varepsilon\mathbf{1}_{[0,t_0)}\mathbf{1}_{\{C=\mathsf{A}\}}$ in the seeded region of Theorem 3.3. At every possible limit history, $a_t \geq a_0 e^{-\delta t}$ since the adopter inflow is nonnegative. Consequently the first-discovery hazard before t_0 is bounded below by $\lambda_M \varepsilon a_0 e^{-\delta t} > 0$. The common process has both a positive probability of no proposal on $[0,T]$ and a positive probability of exactly one accepted proposal in an early subinterval.

Let $H(Y) = \int_0^T e^{-\beta t} R X_t\,\mathrm{d}t$ for a limiting tag. On no proposals $m(\Pi^0) := \mathbb{E}[H(Y) \mid \mathcal{F}^0_T]$ equals the deterministic baseline reward. On exactly one early accepted proposal at time $t \leq t_0$, its value exceeds baseline by at least $e^{-\beta t} L_* m_* e^{-c_0 t}$: before the proposal the baseline still has an incomplete-unexposed fraction at least $m_* e^{-c_0 t}$. Exactly one proposal in an early subinterval and no others has positive probability, with acceptance probability at least $\lambda_M \varepsilon a_0 e^{-\delta t}/L > 0$. Thus m is nonconstant. Conditional independence yields

$$\mathrm{Cov}(H(Y_1), H(Y_2)) = \mathrm{Var}(m(\Pi^0)) > 0.$$

This establishes nontrivial unconditional dependence of reward path functionals, not merely randomness of an unused board. No assertion about the sign of every instantaneous state covariance is needed.

4.4 Scope of the convergence results

The preceding theorem is convergence under a fixed policy and hence, for this bounded payoff and Lipschitz feedback, convergence of that policy's welfare. Equilibrium values converge to the

no-exploration values by (27), and unscaled equilibrium efforts converge uniformly to zero by (17). These are separate results. No convergence of Nu_i^{NE}, of optimal values, or of optimal finite-N controls to a limiting optimizer is claimed.

The secondary, formal limiting planner equation is

$$\begin{aligned} 0 = {} & \partial_t W + F(\mu, b) \cdot D_\mu W + [-\kappa b + w\eta\lambda_w a(\mu)]\partial_b W - \beta W + \langle \mu, g \rangle \\ & + a(\mu) \max_{0 \le u \le \bar{u}} \{\lambda_M u[W(t, \mu, b + h) - W(t, \mu, b)] - c(u)\}, \qquad W(T) = 0. \end{aligned} \tag{31}$$

The disappearance of the private $1/N$ factor is already exact in the finite average-welfare HJB: marginal cost and event productivity both carry $1/N$. Classical differentiability, optimal-policy convergence, and a limiting control verification theorem for (31) remain separate obligations.

5 Creator credit: implementation and its boundary

5.1 Exact two-state implementation

At each discovery, attribute the success to one creator. Pay that creator Np and charge every player p, including the creator. If j creates, player i's net transfer is

$$T_i^j = p(N\mathbf{1}_{\{i=j\}} - 1), \qquad \sum_i T_i^j = 0.$$

This is exact event-by-event balance, not merely an expectation statement. If rivals use u and player i deviates to x in state zero, its discovery and transfer contribution to the Bellman equation is

$$\begin{aligned} & a_N x[D_i + (N-1)p] + a_N(N-1)u[D_i - p] \\ & \quad = a_N[x + (N-1)u]D_i + a_N(N-1)p(x-u). \end{aligned} \tag{32}$$

Thus

$$\beta V_{i,0} = \max_{0 \le x \le \bar{u}} \{-c(x) + a_N x[D_i + (N-1)p] + a_N(N-1)u[D_i - p]\}, \qquad \beta V_{i,1} = v - dD_i.$$

The deviator does not change the transfer charged when a rival creates.

Proposition 5.1 (Balanced first-best credit in the benchmark)**.** *For either productivity normalization, either cost specification, and any cap, let D_{SP} be the corresponding two-state planner value gap. Setting $p = D_{\mathrm{SP}}$ implements the planner effort as a symmetric stationary Markov equilibrium of the transfer game. This is an implementation result, not a claim of uniqueness among all equilibria of that game.*

Proof. At equal efforts, each player's creator intensity is the same, and the expected instantaneous net transfer is zero. Its value therefore equals the planner's average value, giving $D_i = D_{\mathrm{SP}}$. By (32) the maximizing effort is

$$\mathrm{clip}_{[0,\bar{u}]}\left(\frac{a_N[D_{\mathrm{SP}} + (N-1)p] - \ell}{\gamma}\right) = \mathrm{clip}_{[0,\bar{u}]}\left(\frac{Na_N D_{\mathrm{SP}} - \ell}{\gamma}\right) = u_{\mathrm{SP}}.$$

The Bellman equations hold with the planner values; discounted jump compensation verifies the maximizer against all admissible deviations exactly as in Proposition 1.1. For each fixed N, bounded efforts and bounded event payments give integrable discounted transfers. The proof includes zero and cap-bound choices. At $N = 1$ the net transfer is zero. □

The result assumes risk neutrality, reliable success attribution, enforceable levies even on noncreators, commitment to future payment rules, and participation in the funding institution. Free exit before a levy, fake or misattributed artifacts, limited liability, or risk-bearing and verification costs change the game. Transfers cancel from welfare only when those additional costs are absent. The gross prize is order N, financing is order one per person per event, and the private marginal incentive is order one under $a_N = \lambda/N$. An order-one gross creator prize alone supplies only an order-$1/N$ direct marginal incentive.

5.2 Rare large payments and expectations

For the active normalized planner profile, the system discovery process has an N-independent law. Assume $T > 0$ and either an absent initial state, or a present initial state with $d > 0$, so $\mathbb{E}J_T > 0$. For a fixed horizon let C_i^N be player i's attributed success count. Conditional symmetry gives $\mathbb{E}C_i^N = \mathbb{E}J_T/N$ and $\mathbb{P}(C_i^N \geq 1) \leq \mathbb{E}J_T/N$. Thus the gross payment NpC_i^N tends to zero in probability but has expectation $p\mathbb{E}J_T > 0$. It is not uniformly integrable. With discounting, the same argument applies to $Np\int_0^T e^{-\beta t}\,\mathrm{d}C_i^N(t)$.

Under a common construction of the system events and uniformly assigned creator marks, the net tagged payment converges in probability to $-pJ_T$, although its finite-N expectation is zero. State-path weak convergence therefore does not preserve expected transfer payoffs. A continuum mechanism must retain compensators or a marked reward accounting with a separate expectation argument; uniform integrability of raw tagged payments is false in this example, not just an unverified technical condition. Finite-N compensator calculations suffice for the implementation theorem.

Even when expected transfer values can be bounded uniformly using their intensities, the no-transfer lemma cannot be reused: a creator's Hamiltonian contains the immediate $(N-1)p$ payment in addition to the bounded board-value difference.

5.3 A precise six-state criterion

Fix a bounded Borel symmetric target policy with common eligible effort $v(t, \mu, b)$, and let $p(t, \mu, b)$ be a bounded Borel per-event levy. Retain creator identity only as a contemporaneous payment mark. At the target, the instantaneous expected transfer to player i is

$$\lambda_M vp\,(\mathbf{1}_{\{C_i=\mathsf{A}\}} - a(\mu)). \tag{33}$$

This is generally nonzero. Let U_i^p be the actual continuation payoff under this policy and transfer rule. It is determined by the linear, verified evaluation equation

$$(\partial_t + \mathcal{L}_N^v - \beta)U_i^p + g(y_i) - \mathbf{1}_{\{C_i=\mathsf{A}\}}c(v) + \lambda_M vp(\mathbf{1}_{\{C_i=\mathsf{A}\}} - a) = 0, \qquad U_i^p(T) = 0,$$

interpreted mildly. For finite N, bounded transfers and intensity give this well-defined bounded potential.

Proposition 5.2 (Conditional implementation criterion)**.** *The target is a verified Markov equilibrium if, at every eligible state,*

$$v = \mathrm{clip}_{[0,\bar{u}]}\left(\frac{(\lambda_M/N)[\Delta_h U_i^p + (N-1)p] - \ell}{\gamma}\right). \tag{34}$$

For merely Borel rules, necessity holds almost everywhere in time along the target laws from each starting time and state (not only the given initial law); changes at a single time do not alter a payoff.

For continuous rules and continuous evaluated values the condition is necessary pointwise at every preterminal state, by a short-time deviation from that state. If the target is an interior planner optimum, it requires $\Delta_h U_i^p + (N-1)p = N\Delta_h W_N^{\mathrm{SP}}$. Consequently the naive choice $p = \Delta_h W_N^{\mathrm{SP}}$ requires $\Delta_h U_i^p = \Delta_h W_N^{\mathrm{SP}}$ for every eligible player, which does not follow from budget balance or exchangeability.

Proof. For a unilateral deviation the direct event reward has exactly the algebra in (32), with $a_N = \lambda_M/N$, eligibility, and state-dependent p. Rivals retain their feedback rules and transfer evaluation is contained in U_i^p. Strict concavity and the bounded-potential verification identity prove sufficiency. For Borel rules, the evaluation and comparison identities can be obtained directly by conditioning at an independent constant-rate proposal clock: use the baseline killed kernel up to the first proposal, then the accepted/rejected continuation. Iteration is justified by bounded payoffs and the Poisson tail. Thus this version does not assume continuity of a Borel potential for optional stopping. For necessity, mix the target action equally with the measurable clipped maximizer. Concavity gives a nonnegative residual, strictly positive wherever the criterion fails. Actual rare intensity is at least half target intensity; bounded-rate likelihoods make the target law absolutely continuous with respect to this deviating law. Positive target occupation of a strict residual would give a profitable deviation. Necessity follows from every restart; continuity strengthens it to pointwise necessity before T. Budget balance implies $N^{-1}\sum_i U_i^p = W_N^v$, an average identity that does not imply the individual jump identities. □

5.4 An exact counterexample to the naive lift

Take $N = 2$, rare-only mode, $\ell = 0$, and

$$B_0 = 0, \quad Y_1(0) = (1, \mathsf{A}), \quad Y_2(0) = (0, \mathsf{U}), \quad \delta = \nu = r = c_0 = 0, \quad R, \rho, \alpha, c_1, h, \lambda_M > 0.$$

Agent 1 remains a completed adopter forever. Agent 2 proceeds only $\mathsf{U} \to \mathsf{E} \to \mathsf{A}$ and then completes its task. Choose $T > 0$ small enough that $\lambda_M K_T/\gamma < \bar{u}$. The finite planner's feedback is then unbound by the cap. Let $W = W_2^{\mathrm{SP}}$ and propose the naive rule $p = \Delta_h W$ at its optimal profile $v = \lambda_M(\Delta_h W)_+/\gamma$.

Once agent 2 is exposed, the board cannot affect any further reward, so the planner sets all efforts to zero and $p = 0$. While agent 2 is unexposed, only agent 1 is eligible. Whenever $v > 0$, $p = \gamma v/\lambda_M$, and the target's cost and transfer rate for agent 1 cancel *exactly*:

$$-\frac{\gamma v^2}{2} + \frac{\lambda_M}{2} v(2-1)p = 0.$$

When $v = 0$, both terms vanish. Thus in every state of this invariant class $U_1^p(t,s) = K_{T-t}$, independently of the board, and $\Delta_h U_1^p = 0$. At an active target state, using this *target continuation value*, the effort-dependent Hamiltonian is

$$H(x) = \frac{\lambda_M p}{2} x - \frac{\gamma x^2}{2} = \frac{\gamma v x}{2} - \frac{\gamma x^2}{2}, \qquad H(v/2) - H(v) = \frac{\gamma v^2}{8} > 0.$$

This calculation is not a solution of the full dynamically optimal best-response problem. To establish its payoff implication, from (t,s) use $v(r, S_{r-})/2$ for $r \in [t, t+\tau]$ and revert to the target thereafter. Agent 1 remains eligible. The evaluated-potential comparison with transfers gives exactly

$$J_1^{\mathrm{dev}}(t,s) - U_1^p(t,s) = \mathbb{E}_{t,s}^{\mathrm{dev}} \int_t^{t+\tau} e^{-\beta(r-t)} \frac{\gamma v(r, S_{r-})^2}{8}\, \mathrm{d}r = \tau \frac{\gamma v(t,s)^2}{8} + o(\tau) > 0$$

for sufficiently small $0 < \tau < T - t$. The last equality follows from continuity of the planner feedback, stochastic continuity from the finite initial board, and bounded domination. More generally the same residual is nonnegative when the halved policy is used up to a stopping time. There is no gain from changing an action at one isolated time.

There is positive expected occupation of active states under the target from the displayed zero-board initialization. Otherwise the target effort and its artifact intensity would vanish $\mathrm{d}t\ \mathrm{d}\mathbb{P}$-almost everywhere, so its welfare could not exceed zero exploration, contradicting Theorem 3.3 with $a_* = m_* = 1/2$. Dividing $[0, T]$ into finitely many intervals shows that at least one interval has positive expected integral of v^2. Halve the target on that interval, after following it up to the interval's start, then revert. The deviating and target path laws are equivalent on this finite interval: accepted rare intensities have ratio $1/2$ wherever nonzero (and are both zero elsewhere), and all other transition rules coincide. The finite-count likelihood has a strictly positive density. Thus the residual integral is strictly positive under the deviation too. This proves failure of implementation from the stated initialization, as well as failure of Markov perfection at active states.

The counterexample establishes a profitable short-time deviation to half the prescribed effort, not a complete best-response policy. It retains homogeneous costs, perfect attribution, exact budget balance and a verified planner. Existence of individualized funding satisfying (34) is not established here.

6 Contribution and its economic boundary

The economic comparison builds on dynamic public-good and team-production models. Its specific contribution is to connect private production incentives to the limiting shared information environment, with a productive coordinated policy as the welfare comparison. The probability and verification arguments use established tools specialized to the present production institution.

Discovery, durability, and access. Bonatti and Hörner (2011, Sections II–III) already establish free riding and delay in team discovery with uncertainty and hidden effort. Their known-viability case (footnote 11) has individual effort of order $1/n$ without vanishing total effort. Declining individual effort alone is therefore not the contribution. Keller and Rady (2010, Sections 2–5) study genuine Bayesian experimentation through Poisson bandits; our known productivity and persistent usable board have no posterior learning.

Battaglini et al. (2014, Sections I–III) study a depreciating public stock with fixed aggregate resources W and individual endowments W/n. Their resource normalization differs from our fixed physical readiness cost and shrinking access probability. It supplies no empirical justification for uniform opportunity assignment. Battaglini and Palfrey (2024, Section 5, Theorems 4–5), a working paper revised in June 2026, show large-population efficiency or failure depending on how a participation threshold scales; their technology and private-information model differ from ours.

The broader connection between allocation of authority and private initiative is established, for example by Aghion and Tirole (1997). Proposition 1.2 is a narrower calculation about exogenous access probabilities; it does not solve their organizational choice problem. It demonstrates that a fixed opportunity budget alone is insufficient. Uniform dispersed access and pre-assignment readiness costs do the additional economic work.

Probability and verification tools. Hernández-Hernández and Ricalde-Guerrero (2025, Assumptions 1–2, Theorems 7 and 9) develop conditional McKean–Vlasov limits with common marked Poisson noise, using a product noise construction and regularity and moment assumptions. Our

common-proposal coupling adapts established machinery. The integrated width of a thinning mismatch, Poisson domination, and finite-state martingale bounds are elementary; no verbatim embedding of the unbounded exposure model into their full theorem is asserted.

Neumann and Seifried (2026, Theorem 3.11) prove short-horizon existence for a coupled mean-field forward–backward system under Lipschitz reduced coefficients and aggregate-independent relocation. That published result solves a different problem from the finite-N backward construction here. The kernel semigroup perturbation theorem of Kühn and Kunze (2022, Hypothesis 3.1, Theorem 3.2) requires a strong Feller baseline. Our rare-disabled board has deterministic flow in rare-only mode and need not smooth Borel functions, so that theorem cannot be imported directly. Appendix A.1 proves the potential and interlacing identities under this model's actual clock construction. Banach contraction and reward-bound continuation are proof tools, not new general game-existence machinery.

Delarue and Foguen Tchuendom (2020, Section 6, Theorem 19) study randomized selection among deterministic mean-field equilibria. Their finite-player discussion analyzes first-order systems and explicitly distinguishes them from establishing Nash existence. This precedent motivates caution about broad novelty claims for incentives and limiting randomness; it is not an existence theorem for our game. Holmström (1982, Section 2) studies budget-balanced output sharing in teams. Attributed creator prizes use extra creator information and enforceable levies; they do not overturn an output-only sharing impossibility.

The connected result. The analysis supplies a finite-population equilibrium, uniform control of its path law and welfare, a productive fixed policy with an N-uniform gain, and strictly positive covariance of that policy's limiting tagged task rewards. The covariance statement ensures that the retained common history affects useful outcomes. Creator credit adds a boundary: budget balance alone does not transport a homogeneous benchmark implementation to heterogeneous private states. These are model-specific results, not claims of a general mechanism or of novelty from combining public-good and common-noise ideas.

7 Numerical methods and evidence

The calculations use a two-state policy evaluator and a six-state controlled piecewise deterministic simulator. A fixed configuration and independently seeded run records underlie the tables and figure.

The benchmark is checked against direct 2×2 linear policy evaluation, bounded scalar best-response optimization, planner policy optimization, occupancy ODE integration, and the convolution formula for first discovery from state one. The tests include $N = 1$, zero productivity and benefits, $d = 0$, positive linear costs, fixed productivity, and binding effort caps. They verify the proposed credit rule against a deviating policy rather than only substituting the same equilibrium formula twice.

The controlled simulator aggregates the 14 unchanged individual transition rates, retaining labels for state changes and creator attribution. Its thinning envelope is the current sum of individual and routine rates plus the constant rare bound $\lambda_M \bar{u}$. The first part decreases along the intervening board flow; the last part remains valid when effort rises with time or as the board decays. The candidate-time acceptance rate uses the decayed board and left-limit private state. Tests explicitly start a time-increasing policy at zero current intensity, a case in which reusing the old total rate would fail. Generator tests verify recovery of the original mechanism at $u = 1$ and that $u = 0$ disables only rare production. Other checks cover the seed obstruction, simplex and board invariance,

creator indices, and exact transfer balance.

Rewards are integrated exactly between jumps, including discounting. Constant early effort has exact integrated effort, cost, and hazard formulas; the two smooth diagnostic feedbacks use adaptive quadrature. Initial label permutation and dynamic events use independent child streams of the recorded SeedSequence. No board grid, Euler time step, or board truncation is used by the finite simulator. The fixed-policy limit uses exact Poisson proposal times and numerical ODE solutions between them, split at the policy's deterministic stopping time.

Table 1: Normalized quadratic benchmark, $\lambda = \beta = d = v = \gamma = 1$, $\ell = 0$ and $\bar{u} = 2$.

N	Nash effort	Planner effort	Nash value at 0	Nash mean wait
1	0.449490	0.449490	0.101021	2.2247
10	0.048866	0.449490	0.022685	20.4642
100	0.004988	0.449490	0.002475	200.4963
1000	0.000500	0.449490	0.000250	2000.4996

Table 2: Fixed coordinated early effort: independent-history means and 95% intervals.

N	Welfare above baseline	Mean artifacts	Integrated intensity
60	0.1147 [0.0469, 0.1825]	0.1125	0.1158
240	0.1947 [0.1184, 0.2710]	0.1750	0.1253
960	0.1240 [0.0618, 0.1863]	0.1062	0.1186
Limit	0.0998 [0.0442, 0.1554]	0.0750	0.1174

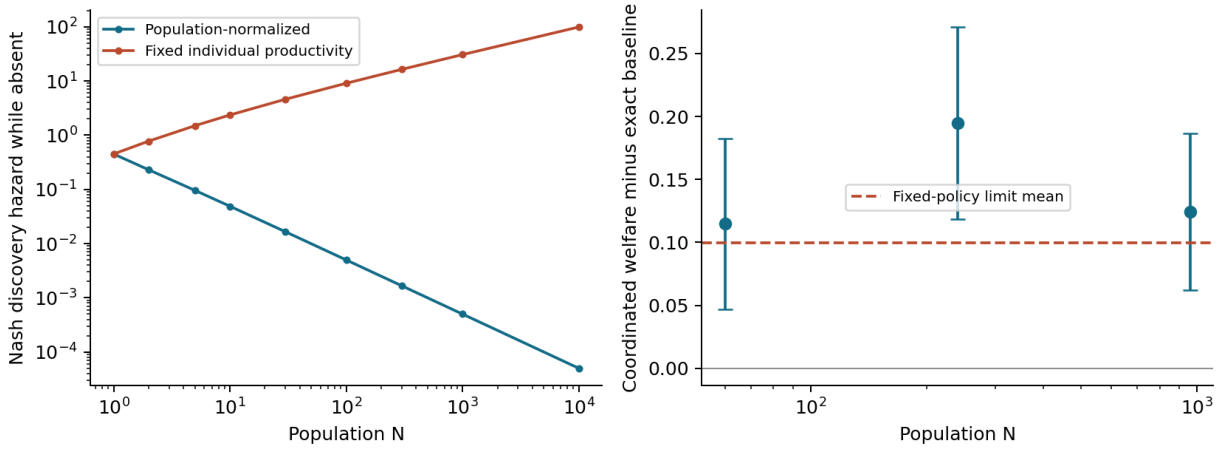


Figure 1: Left: the two-state equilibrium hazard under the two productivity normalizations. Right: the six-state fixed coordinated policy's welfare above its exact no-exploration baseline. Error bars use independent common histories.

The full-model experiment has $T = 6$, $\beta = 0.1$, $R = 1$, $\gamma = 0.1$, $\ell = 0$, $\bar{u} = 1$, and eligible effort 0.25 until time 1.25. Initial counts are 20% adopters and 80% incomplete-unexposed, all tasks initially incomplete. Parameters are $\lambda_M = 2$, $h = 2$, $\rho = \alpha = 2$, $\kappa = 0.4$, $\nu = 0.15$, $\delta = r = 0.2$, $c_0 = 0.1$, $c_1 = 1.2$. The routine channel is off. Each cell uses 160 independent histories at $N = 60, 240, 960$;

the limit has 160 independent histories. The seed root is 20260909, and complete seed formulas and per-run seeds are saved.

The exact no-exploration expected welfare is approximately 1.15065544. The separately computed first-order coefficients are approximately 0.576314, 0.576926, and 0.577079 at the three population sizes. These numerical coefficients use (36) and ODE/quadrature calculations, not simulation. The explicit theorem witness instead gives the deliberately conservative $A_* \simeq 9.61\,10^{-4}$, a certified effort of about $3.70\,10^{-7}$ and a certified gap of about $1.78\,10^{-10}$. The illustrated effort 0.25 is not certified by this coarse constant. Its larger welfare improvement is numerical evidence; the mathematical theorem certifies the existence of some fixed positive improvement.

The saved experiments also include zero effort and an imposed $0.25/N$ early-effort diagnostic. The latter is *not* a computed Nash equilibrium. Conditional hazards, realized counts, integrated intensities, task reward, average cost, and baseline-adjusted welfare are recorded separately. Student intervals use histories as independent replications; probabilities use Wilson intervals. They are finite-sample diagnostics with only 160 histories, not finely resolved estimates of extremely rare events. The artifact counts and integrated hazards are recorded independently, permitting a compensator check. Dependence of limiting tagged rewards is proved in Section 4; the sample variance of conditional mean reward is only an illustrative numerical diagnostic.

For this release, the numerical test suite was run afresh, deterministic coefficients and the witness constants were recomputed, and saved histories were checked against their reported aggregates. The displayed 1,440 finite and 160 limiting histories were retained, not resimulated. Numerical tests check implementation and numerical identities; they do not prove equilibrium existence or convergence. No numerical six-state Nash solve was performed.

The local editable release includes the complete LaTeX, bibliography, figure, tables and a build script. The public posting contains the paper and accompanying summary PDFs. Simulation code and run records remain in the local research project and are not part of this public release. The full parameter values, policy, replication counts and numerical methods are specified above; independent reproduction of the simulations from a public code package remains unavailable with this version.

8 Interpretation and next question

Under uniformly allocated, nontransferable opportunities and readiness paid before assignment, private incentives suppress the rare information channel as the population grows. The finite-population equilibrium and coupling bounds make this conclusion explicit. In the seeded, zero-board, rare-only productive region, a specified coordinated policy yields a positive per-capita gain and retains common jumps that induce unconditional dependence in tagged rewards.

The assumptions delimit the comparison. Equilibrium uniqueness is restricted to the bounded jointly continuous mild-HJB class. The welfare positivity result does not cover arbitrary initial boards or hybrid backgrounds. Fixed-policy convergence does not imply convergence of optimal values or controls, and the general controlled hybrid argument is qualitative. The productivity-scaling classification and first-best creator-credit implementation belong to the two-state benchmark. In the six-state counterexample, a positive-duration half-effort deviation followed by reversion disproves the naive implementation; a complete best-response policy and a general heterogeneous-state mechanism are not established.

The economic limitation is the access institution. Fixed capacity by itself does not force collapse, as designated access in the benchmark demonstrates. The paper supplies no evidence that uniform non-reassignable access and pre-assignment readiness describe a particular AI-agent system. Productivity is known, states are observed, and the scalar board omits content, permissions, and

artifact-specific exposure. No Bayesian learning, endogenous hierarchy, efficient specialization, or empirical causal claim follows from the model.

A bounded next question is how reassignment of unused opportunities changes the comparison while holding the physical opportunity rate and task transitions fixed. Specifying that alternative allocation rule and the timing of readiness costs would test the economic premise directly. The present release records the conditional comparison at this stage of the research.

A Verification and supporting estimates

A.1 Potential verification under predictable interlacing

All semigroup integrals here are pointwise bounded kernel integrals. No Bochner integral or global strong continuity on C_b is required. On a terminal slab $[a, b]$, define

$$Q_\xi \psi(t) = e^{-\xi(b-t)} P^0_{b-t}\psi, \qquad R_\xi f(t) = \int_t^b e^{-\xi(r-t)} P^0_{r-t} f(r)\,\mathrm{d}r.$$

The semigroup property and bounded Fubini give $Q_\beta - Q_{\beta+L} = LR_{\beta+L}Q_\beta$ and $R_\beta - R_{\beta+L} = LR_{\beta+L}R_\beta$. Thus $V = Q_\beta\psi + R_\beta F$ solves $V = Q_{\beta+L}\psi + R_{\beta+L}(F+LV)$. Conversely, for fixed bounded F the latter linear Volterra equation has a unique bounded solution: its homogeneous difference satisfies $\|D(t)\| \le L\int_t^b \|D(r)\|\,\mathrm{d}r$ and vanishes by iteration. Taking $F = g - c(A[V]) + \Lambda[V]\Delta_h V$ proves killed/un-killed equivalence even though this F was originally defined through V.

Lemma A.1 (Controlled potential identity)**.** *Let V, F be bounded jointly continuous functions satisfying $V = Q_\beta\psi + R_\beta F$. Retain baseline private and routine transitions, and add $s \mapsto s^h$ jumps with any predictable intensity $0 \le \lambda_r \le L$. For every bounded stopping time $\sigma \in [t, b]$,*

$$V(t,s) = \mathbb{E}_{t,s}\left[e^{-\beta(\sigma-t)}V(\sigma, S_\sigma) + \int_t^\sigma e^{-\beta(r-t)}\{F(r, S_{r-}) - \lambda_r \Delta_h V(r, S_{r-})\}\,\mathrm{d}r\right]. \tag{35}$$

The associated process is a true martingale for the admissible filtration.

Proof. Splitting the mild identity at deterministic r shows that under a baseline restart from (t, s),

$$Z^0_r = e^{-\beta(r-t)}V(r, S^0_r) + \int_t^r e^{-\beta(v-t)}F(v, S^0_v)\,\mathrm{d}v$$

is the conditional expectation of the bounded terminal variable $e^{-\beta(b-t)}\psi(S^0_b) + \int_t^b e^{-\beta(v-t)}F(v, S^0_v)\,\mathrm{d}v$. It is uniformly integrable and cadlag, by joint continuity of V and cadlag baseline paths. Private and routine future increments remain independent of the past when the filtration includes unused past proposal marks and independent control randomization, so this martingale property persists in the enlarged admissible filtration.

For $L > 0$, take independent rate-L proposal times σ_k with independent uniform marks U_k. Accept at σ_k if $U_k \le \lambda_{\sigma_k}/L$, evaluating the intensity before revealing U_k. After each proposal restart the baseline from the actual post-proposal state, using fresh future private and routine increments. Random restart times and states are allowed by the baseline strong Markov property. Until the next proposal the path is baseline. Optional stopping of its bounded potential martingale at that independent exponential time, cut off at b, gives a zero conditional mean segment increment. Predictable deviations can depend on the whole past; they affect acceptance and do not change the law of this baseline segment before its next proposal.

Sum the segment martingales, initially stopping after m proposals. There are almost surely no simultaneous private/routine and rare proposals. An accepted proposal adds exactly $e^{-\beta(\sigma_k-t)}\Delta_h V(\sigma_k, S_{\sigma_k-})$; a rejected one adds zero. Hence

$$e^{-\beta(r-t)}V(r,S_r) + \int_t^r e^{-\beta(v-t)}F(v,S_{v-})\,\mathrm{d}v \\ - \sum_{\sigma_k\le r}^{\text{accepted}} e^{-\beta(\sigma_k-t)}\Delta_h V(\sigma_k, S_{\sigma_k-})$$

is a stopped martingale. Its absolute value is bounded by a constant times $1+P_L(b-t)$, where P_L is the rate-L proposal count. This integrable domination permits $m\to\infty$ in conditional expectations.

Predictable thinning gives compensator $\lambda_v\,\mathrm{d}v$. Subtract from the displayed jump sum its compensator

$$\int_t^r e^{-\beta(v-t)}\lambda_v\Delta_h V(v,S_{v-})\,\mathrm{d}v.$$

The result is a true martingale: the integrand is bounded by $2\|V\|_\infty$ and the expected count by $L(b-t)$. Adding that compensated martingale to the preceding one gives the claimed martingale. Its expression is bounded by

$$\|V\|_\infty + (b-t)(\|F\|_\infty + 2L\|V\|_\infty).$$

Optional stopping gives (35). For $L=0$, the baseline identity suffices. □

Under a unilateral deviation, rivals retain $A_j[V](r,S_{r-})$, and

$$\lambda_r = \frac{\lambda_M}{N}\left(u_i(r) + \sum_{j\neq i} A_j[V](r,S_{r-})\right).$$

Rival actions and values are evaluated at the deviating state. The lemma with $\sigma=b$ gives the main proof's Hamiltonian residual and includes all future feedback responses. Applied to the rare-disabled welfare potential, it also proves (21) for a Borel gate: the gate enters only the bounded predictable actual intensity, not the continuous V or F.

A.2 A precise numerical formula for the one-artifact coefficient

In rare-only mode at zero board and zero effort, labeled chains are independent. Let $p_j(t,\cdot)$ be the distribution of a chain initially in state j, f_j the initial deterministic fraction in that state, and

$$d_y(t) = w_y(T-t,h) - w_y(T-t,0),$$

where $w_y(H,b)$ is the remaining discounted reward of a single chain with deterministic decaying board $be^{-\kappa r}$ and no further deposits. Then $\Delta_h W_N^0 = \sum_y \mu^N(y)d_y$, and writing $\bar a = \sum_j f_j p_j(\mathsf{A})$ and $\bar d = \sum_j f_j \sum_y p_j(y)d_y$ gives

$$\mathbb{E}^0[a(\mu_t^N)\Delta_h W_N^0] = \bar a\bar d + \frac{1}{N}\sum_j f_j\left[\sum_{y:C(y)=\mathsf{A}} p_j(y)d_y - p_j(\mathsf{A})\sum_y p_j(y)d_y\right]. \tag{36}$$

To derive it, expand the product of the two empirical averages. Distinct labels' terms factor; equal-label terms contribute exactly the displayed covariance. This retains the $1/N$ correction rather than replacing an expected product by a product of expectations.

The code computes $p_j(t)$ using the matrix exponential of Q_0. For w_y, it solves $P'(r) = P(r)Q_{be^{-\kappa r}}$, $P(0) = I$, together with $z'(r) = e^{-\beta r}P(r)g$, $z(0) = 0$, and returns $z(H)$. At $b = 0$ this is checked against the independent matrix-resolvent identity

$$z(H) = (Q_0 - \beta I)^{-1}(e^{(Q_0-\beta I)H} - I)g$$

when $\beta > 0$. Gaussian quadrature of (36) gives A_N; quadrature-order refinement is a numerical check, not a proof of positivity.

A.3 Stopping, martingale moments, and regularity boundaries

For any bounded policy and $p > 2$, the empirical martingale has coordinate jump sizes at most $1/N$ and predictable bracket at most $CN^{-1}\int_0^T(1 + B_s^N)\,\mathrm{d}s$. The jump version of the Burkholder–Davis–Gundy inequality bounds its pth supremum moment by a constant times the $p/2$ bracket moment plus $CN^{1-p}\mathbb{E}\int_0^T(1+B_s^N)\,\mathrm{d}s$. Since $N^{1-p} \le N^{-p/2}$ for $p \ge 2$, (12) yields $\|\sup|M^N|\|_{L^p} \le C_T N^{-1/2}$. The routine-board martingale has the same estimate from rate $N\lambda_w a$ and jump η/N.

For the continuous-board equilibrium proof, the semigroup argument operates on continuous bounded functions, not on a putative bounded classical generator. The potential's martingale identity supplies the needed extended generator property; it is precisely why a differentiability estimate is unnecessary. The local contraction cannot alone prove global existence: verification and the uniform reward bound are the continuation step. Conversely, compact controls and strict concavity alone would not solve a general coupled game lacking the explicit maximizing selector in (13).

A.4 The rare-disabled population limit

Here are the details of the baseline estimate used in Section 4, also underlying the nonstrategic weak limit of Sneller (2026). Under the initialization conditions stated there, the vector field in (28) is locally Lipschitz, conserves population mass, and points inward on the simplex and at board zero. Since $a \le 1$, its board is bounded by $b_0 + w\eta\lambda_w T$. This proves existence and uniqueness on $[0, T]$.

Random time change for the private transitions and routine posts gives

$$\mu_t^{N,0} = \mu_0^N + \int_0^t F(\mu_s^{N,0}, B_s^{N,0})\,\mathrm{d}s + M_t^{N,\mu},$$

$$B_t^{N,0} = B_0^N + \int_0^t [-\kappa B_s^{N,0} + w\eta\lambda_w a(\mu_s^{N,0})]\,\mathrm{d}s + wM_t^{N,w}.$$

The moment calculation above bounds both martingale suprema in L^p by $C_T N^{-1/2}$. Let Z_N be their summed suprema plus the initial error, and let $D_t = \sup_{s\le t}(\|\mu_s^{N,0} - \mu_s\|_1 + |B_s^{N,0} - b_s|)$. The exposure product bound in Section 4 gives pathwise

$$D_t \le Z_N + C\int_0^t (1 + B_s^{N,0})D_s\,\mathrm{d}s, \qquad D_T \le Z_N \exp\left(C\int_0^T (1 + B_s^{N,0})\,\mathrm{d}s\right).$$

Domination by $K + w\eta P_w(N\lambda_w T)/N$ supplies a uniform exponential moment of the last factor at every fixed power. Hölder's inequality with exponents p and $p/(p-1)$ therefore yields $\mathbb{E}D_T \le C_T(\varepsilon_{N,p} + N^{-1/2})$. This proves the stated expected uniform estimate without an unproved global Lipschitz bound for the board. The tagged coupling in Section 4 then gives ordinary path-space independence for exchangeable chaotic initial arrays.

References

Philippe Aghion and Jean Tirole. Formal and real authority in organizations. *Journal of Political Economy*, 105(1):1–29, 1997. doi: 10.1086/262063. URL https://www.journals.uchicago.edu/doi/10.1086/262063.

Marco Battaglini and Thomas R. Palfrey. Dynamic collective action and the power of large numbers. Technical Report 32473, National Bureau of Economic Research, 2024. URL https://www.its.caltech.edu/~trp/Dynamic%20Volunteer_09June2026_wp.pdf. Revised June 9, 2026.

Marco Battaglini, Salvatore Nunnari, and Thomas R. Palfrey. Dynamic free riding with irreversible investments. *American Economic Review*, 104(9):2858–2871, 2014. doi: 10.1257/aer.104.9.2858. URL https://snunnari.github.io/bnp_dynfreerider_aer.pdf.

Alessandro Bonatti and Johannes Hörner. Collaborating. *American Economic Review*, 101(2):632–663, 2011. doi: 10.1257/aer.101.2.632. URL https://bpb-us-w2.wpmucdn.com/campuspress.yale.edu/dist/3/352/files/2012/02/65916.pdf.

François Delarue and Rinel Foguen Tchuendom. Selection of equilibria in a linear quadratic mean-field game. *Stochastic Processes and their Applications*, 130(2):1000–1040, 2020. doi: 10.1016/j.spa.2019.04.005. URL https://arxiv.org/abs/1808.09137.

Daniel Hernández-Hernández and Joshué Helí Ricalde-Guerrero. Conditional McKean–Vlasov differential equations with common Poissonian noise: Propagation of chaos. *Stochastics and Dynamics*, 25(7–8):2550033, 2025. doi: 10.1142/S0219493725500339. URL https://arxiv.org/abs/2308.11564. Article 2550033; author version arXiv:2308.11564v3.

Bengt Holmström. Moral hazard in teams. *The Bell Journal of Economics*, 13(2):324–340, 1982. doi: 10.2307/3003457. URL https://www.jstor.org/stable/3003457.

Godfrey Keller and Sven Rady. Strategic experimentation with Poisson bandits. *Theoretical Economics*, 5(2):275–311, 2010. doi: 10.3982/TE595. URL https://users.ox.ac.uk/~econ0138/Research/PB.pdf. Author version dated January 13, 2010.

Franziska Kühn and Markus Kunze. Feller generators with measurable lower order terms. *Positivity*, 26:85, 2022. doi: 10.1007/s11117-022-00948-4. URL https://link.springer.com/article/10.1007/s11117-022-00948-4. Article 85.

Berenice Anne Neumann and Frank T. Seifried. Finite state mean field games with common shocks. *Applied Mathematics & Optimization*, 94:26, 2026. doi: 10.1007/s00245-025-10334-7. URL https://link.springer.com/article/10.1007/s00245-025-10334-7. Article 26.

Lucas Sneller. Persistent information in large agent populations: Weak and rare-artifact limits. Unpublished independent working manuscript, August 31, 2026, 2026.